

## Bhavana Nare

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### PROFESSIONAL SUMMARY

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- Senior Software Engineer with 10+ years of experience designing and deploying AI/ML-enabled platforms, including data pipelines, AI integration automation, and full stack engineering.
- Strong background in ML integration and AI platforms, combining cloud-native architecture, MLOps workflows, and DevOps automation to operationalize ML capabilities at scale.
- Proven impact architecting AI-driven systems that reduced operational risk, cut manual effort by 60–70%, and scaled into enterprise CI/CD environments.
- Expertise in system architecture, performance optimization, and automation, with hands-on leadership across the full SDLC.

### PUBLICATION

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#### A Novel Computational Framework of Robot Trust for Human-Robot Teams - IEEE, 2025

- Developed a computational trust model for robots that quantifies how autonomous agents assess and update trust in human teammates based on interaction outcomes, enabling robots to adapt team behavior dynamically.
- Formulated and implemented a psychological trust-building process into a computational framework, deriving predictors of positive/negative human–robot experiences and representing evolving belief/affinity in multi-agent teams.
- Demonstrated how trust and kinship metrics influence robot decision making in human–robot collaboration and provided a basis for trust-aware autonomous behavior in multi-agent systems.

#### Interpretable Deep Learning for Biological Age Prediction: A Counterfactual Approach to Personalized Health Insights

Atlantis Press, 2026

- Designed an Explainable Time Series Regression (XTSR) framework integrating deep learning with counterfactual reasoning.
- Trained hybrid time-series models on large-scale wearable sensor data to generate interpretable biological age predictions.

#### Wildfire Prediction and Visualization: A Machine Learning Approach Using U.S. Data - Atlantis Press, 2026

- Analyzed ~1.88 million U.S. wildfire incidents using machine learning and data visualization techniques.
- Identified patterns in frequency, distribution, and contributing causes using large-scale structured datasets.

### CERTIFICATION

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- **ISTQB Certified Tester (CTFL)** - ITB, Dec 2020
- **Build a Computer Vision App with Azure Cognitive Service** – Microsoft, Feb 2026

### EDUCATION & RESEARCH

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- **Master of Computer Science (Thesis)** — University of Georgia, Athens, GA  
Aug 2021 – May 2023 | GPA: 3.7 / 4.0
- **Bachelor of Technology in Computer Science** — Sree Vidyanikethan Engineering College, Tirupati, India  
Oct 2010 – Apr 2014 | GPA: 7.9 / 10

### PROFESSIONAL EXPERIENCE

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#### Cybersecurity AI Analyst

#### *Rivian Automotive (May 2025 –Present)*

- Architected and deployed Python tool Beacon, a low-latency AI-powered risk decisioning and automated remediation system operating across ~120 repositories and ~400–600 merge requests per day.
- Designed and optimized a two-stage real-time decisioning pipeline (lightweight ML pre-filter → LLM + RAG deep analysis) achieving ~70% noise reduction while maintaining sub-25s inference latency.
- Implemented production-grade RAG architecture using pgvector embeddings, repository history, CODEOWNERS metadata, and historical exploit patterns to enable context-aware AI risk scoring.
- Reduced inference latency from ~90s to ~18–25s via async orchestration, batched embedding retrieval, and prompt context optimization.
- Delivered automated remediation workflows (MR comments, Slack alerts, policy citations) reducing manual triage effort by ~60–70%.
- Established SLOs, observability, and rollback mechanisms to ensure safe and reliable AI deployment in CI/CD environments.
- Integrated deterministic guardrails (Pydantic schema validation, mandatory policy citation enforcement) to maintain compliance and auditability.

- Built a continuous learning loop via Databricks, re-indexing developer feedback into the RAG pipeline to improve classification accuracy over time.

## **MLOps Engineer**

***Robert Bosch (Aug 2023 –April 2025)***

- Designed and deployed a cloud-native risk signal processing pipeline (EventBridge → Lambda → S3 → DynamoDB) ingesting CI/CD telemetry across ~80+ repositories to generate structured behavioral signals.
- Engineered deterministic and ML-driven risk features (file-diff volatility, failure frequency, integration instability, review latency, historical defect density) to model pull-request risk propensity.
- Developed supervised classification models (logistic regression, tree-based models) for PR risk scoring, categorizing changes into Testing / Pipeline / Software risk domains.
- Implemented rule + model hybrid decision logic combining metadata heuristics with statistical scoring to improve precision and reduce false-positive alerts.
- Built online scoring services using Python to generate near-real-time risk assessments for incoming pull requests.
- Established monitoring pipelines (CloudWatch metrics, structured logging) to track model drift, feature distribution shifts, and scoring stability.
- Delivered risk dashboards visualizing throughput volatility, failure clustering, and release risk concentration, enabling engineering leadership to prioritize high-risk integrations.

## **Senior Software Engineer –Computer Vision**

***Continental Automotive (May 2019 – July 2021)***

- Contributed to ML-based computer vision object detection pipelines for ADAS camera systems, delivering perception outputs consumed by Emergency Braking Assistance (EBA) and other safety-critical downstream functions.
- Integrated and curated labeled training datasets into YOLO-based object detection pipelines, enabling iterative model training, validation, and performance evaluation across multiple data versions and operating conditions.
- Supported the transition from 2D to 3D bounding box detection, improving object localization, depth awareness, and spatial accuracy for perception outputs used in ADAS decision-making logic.
- Collaborated cross-functionally with labeling, software engineering teams to analyze detection behavior, identify failure modes, and refine model outputs against functional requirements.
- Validated perception outputs against accuracy, recall/precision thresholds, and production-readiness criteria, ensuring alignment with ADAS safety and performance expectations.
- Assisted in evaluating model performance across diverse scenarios and edge cases, supporting data-driven decisions for model iteration and deployment readiness.
- Built Python-based data processing pipelines to normalize detection outputs and generate longitudinal performance insights for model regression monitoring.

## **ADDITIONAL EXPERIENCE**

- **Python Developer & Data Analyst** — Teradata India Pvt. Ltd August 2018 – May 2019
- **Senior Software Engineer** — Tata Consultancy Services (TCS) June 2014 – August 2018

## **TECHNICAL SKILLS**

- **Programming & Languages:** Python, Java, C++, SQL, Bash/Shell, JavaScript (React)
- **AI / ML & LLM Platforms:** Vertex AI, AWS Bedrock, PyTorch, scikit-learn, Pandas, NumPy, OpenCV
- **Cloud & MLOps:** AWS (Lambda, SageMaker, S3, DynamoDB, RDS, API Gateway, Step Functions), Google Vertex AI, Docker, Terraform, GitHub Actions, GitLab CI, Jenkins
- **Data & Storage:** DynamoDB, PostgreSQL, Amazon RDS, Databricks, S3, Vector Databases (PG Vector)
- **Observability & Analytics:** CloudWatch, Structured Logging, Metrics & Alerting, GraphQL, Matplotlib, Hex
- **Data Storage & Retrieval:** DynamoDB, PostgreSQL, Amazon RDS, S3, pgvector (Vector Databases)
- **Engineering Practices:** Microservices Architecture, System Design, Performance Optimization, Agile Development